

11.2 部分习题参考简答

2. (1) $\frac{\partial z}{\partial x} = \frac{y(x^2 - y^2)}{x^4 + 3x^2y^2 + y^4}, \quad \frac{\partial z}{\partial y} = \frac{x(y^2 - x^2)}{x^4 + 3x^2y^2 + y^4};$
- (2) $\frac{\partial z}{\partial x} = 2xf_1' + ye^{xy}f_2', \quad \frac{\partial z}{\partial y} = -2yf_1' + xe^{xy}f_2';$
- (3) $\frac{\partial z}{\partial x} = f_1' + f_2' + f_3', \quad \frac{\partial z}{\partial y} = f_2' - f_3';$
- (4) $\frac{\partial z}{\partial x} = y - \frac{y}{x^2}f(xy) + \frac{y^2}{x}f'(xy), \quad \frac{\partial z}{\partial y} = x + \frac{1}{x}f(xy) + yf'(xy);$
- (5) $\frac{\partial z}{\partial x} = (1 + \ln x)f_1' + 2f_2', \quad \frac{\partial z}{\partial y} = -f_2'.$

9. (1) $f_{xx}''(0,0) = f_{xy}''(0,0) = 0.$

(2) $f_{x^n y^m}^{(n+m)}(0,0) = \begin{cases} 0, & n = 2k, \\ (-1)^{k-1}, & n = 2k-1, \end{cases} \quad \text{其中 } k \in \mathbb{N}^+.$

10. (1) $f_{xx}'' = -2a^2 \cos(2ax - 2by), \quad f_{xy}'' = 2ab \cos(2ax - 2by),$

$$f_{yy}'' = -2b^2 \cos(2ax - 2by);$$

(2) $f_{xx}'' = a^2 e^{-ax} \sin by, \quad f_{xy}'' = -abe^{-ax} \cos by, \quad f_{yy}'' = -b^2 e^{-ax} \sin by;$

(3) $f_{xx}'' = (-2y + xy^2)e^{-xy}, \quad f_{xy}'' = (-2x + x^2y)e^{-xy}, \quad f_{yy}'' = x^3 e^{-xy};$

(4) $f_{xx}'' = -\frac{1}{(x + \sqrt{1-y^2})^2}, \quad f_{xy}'' = \frac{y}{(x + \sqrt{1-y^2})^2 \sqrt{1-y^2}},$

$$f_{yy}'' = -\frac{x + (1-y^2)^{\frac{3}{2}} + 2y^2 \sqrt{1-y^2}}{(x\sqrt{1-y^2} + 1 - y^2)^2 \sqrt{1-y^2}}.$$